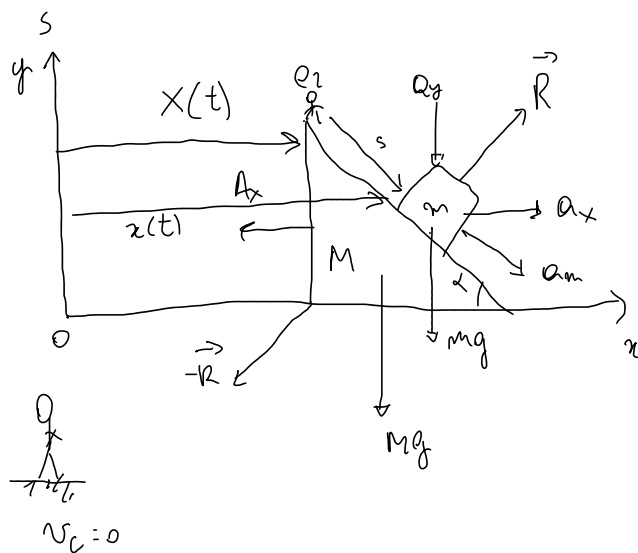




$$a_y, a_x, A_x$$

$$(iii) \boxed{a_y = (A_x - a_x) \tan \alpha}$$

$$(i) a_y = -\ddot{s} \sin \alpha = -a_m \sin \alpha$$

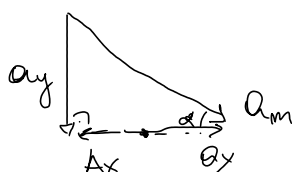
$$(ii) a_x = A_x + a_m \cos \alpha$$

$$(iii) a_m = -\frac{a_y}{\sin \alpha} = \frac{a_x - A_x}{\cos \alpha}$$

↳

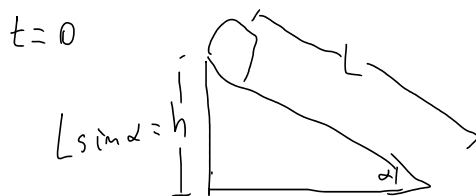
$$\cos \alpha = \frac{a_x - A_x}{a_m}$$

$$\tan \alpha = \frac{a_y}{a_x - A_x}$$



$$x = A_x \frac{t^2}{2}, \quad y = a_y \frac{t^2}{2}, \quad x = a_x \frac{t^2}{2}$$

$$a_x = \frac{Mg \sin \alpha \cos \alpha}{M + m \sin^2 \alpha}, \quad A_x = -\frac{mg \sin \alpha \cos \alpha}{M + m \sin^2 \alpha}, \quad a_y = -\frac{(M+m)g \sin^2 \alpha}{M + m \sin^2 \alpha}$$



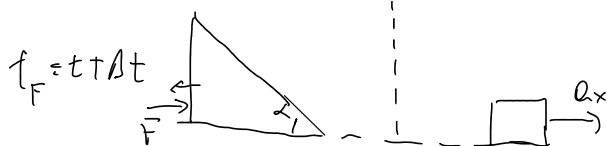
$$E = mgL \sin \alpha \quad (A_y = 0)$$

$$V_x = 0, \quad v_x = v_y = 0, \quad a_x = a_y = 0, \quad A_x = 0$$

$t_f = t$



$$\begin{cases} E = \frac{1}{2} M V_x^2 + \frac{1}{2} m (v_x^2 + v_y^2) + mg(L \sin \alpha - h(t)) \\ h(t) = s(t) \cdot \sin \alpha \end{cases}$$



$$E = \frac{1}{2} M V_x^2 + \frac{1}{2} m v_x^2 = mgL \sin \alpha$$

$$\odot: \vec{p}^{(A)} = \vec{p}^{(0)} \Rightarrow \vec{p} = M \vec{V} + m \vec{v} = 0$$

$$(*) M V_x + m v_x = 0$$

$$| -D_{x \dots} = M A_x \quad | F - R \sin \alpha = M A_x$$

$$(*) \quad M V_x + m v_x = 0$$

$$(**) \quad M A_x + m a_x = 0 \rightarrow \left\{ \begin{array}{l} -R \sin \alpha = M A_x \\ R \sin \alpha = m a_x \end{array} \right. \Rightarrow \left\{ \begin{array}{l} F - R \sin \alpha = M A_x \\ R \sin \alpha = m a_x \end{array} \right.$$

$$V_x = - \frac{m v_x}{M}$$

$$M V_x^2 + m v_x^2 = 2 m g L \sin \alpha$$

$$M \frac{v_x^2}{M^2} + m v_x^2 = 2 m g L \sin \alpha$$

$$\left( \frac{m}{M} + 1 \right) v_x^2 = 2 g L \sin \alpha$$

$$v_x = \sqrt{\frac{2 g L \sin \alpha}{1 + \frac{m}{M}}} \Rightarrow V_x = - \frac{m}{M} v_x = - \frac{m}{M} \sqrt{\frac{2 g L \sin \alpha}{1 + \frac{m}{M}}}$$

$$M \gg m \therefore v_x \xrightarrow{\frac{m}{M} \rightarrow 0} \sqrt{2 g L \sin \alpha}, \quad V_x \xrightarrow{\frac{m}{M} \rightarrow 0} 0$$

$$F = M A_x + m a_x$$

$$\int_0^{\Delta t} F dt = M \int_0^{\Delta t} A_x dt + m \int_0^{\Delta t} a_x dt$$

$$I = M V_x + m v_x$$